Dae Hyun Lee

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Education

University of Washington — M.S. Data Science — Mar 2027 (Expected)
University of Washington — B.A. Mathematics — Mar 2025

Technical Skills

Languages: Python, Java, R, SQL
ML/DL Frameworks: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, Random forest
Machine Learning Skills: End-to-End ML Pipelines, LLMs, ML Operations, Supervised & Unsupervised Learning
Others: Spark, Pandas, NumPy, Matplotlib, Seaborn, Tableau, Dash, Plotly, Google Cloud, Excel, Git

Work Experience

Researcher — Oct 2025 – Current
Roy Lab, University of Washington

- Developed and benchmarked deep learning architectures (3D CNN, ViT, ResNet, ResNeXt) within an end-to-end supervised learning pipeline across ADNI, OASIS-3, and HCP (4,508+ subjects, 127 clinical features) for three-way Alzheimer's classification (CN/MCI/AD), improving accuracy from 55% to 70% toward a generalizable foundation model
- Reframed prediction heterogeneity as a signal of cognitive reserve rather than noise, using patient-level data splitting, class-balanced sampling, and 5-fold CV inference to identify cognitive reserve candidates from structural MRI and clinical biomarkers; targeting precision medicine applications in early-stage neurodegenerative disease
- Leading manuscript preparation for a review paper on cognitive reserve measurement methodologies in neuroimaging (submission pending)

Research Intern — Jun 2024 – Sep 2024
SNU VLDB Lab, Seoul National University

- Proposed and validated a z-score-based dimension reduction technique for embedding vectors, supported by mathematical formulation and statistical analysis
- Achieved 96% dimensionality reduction (1536→64 dimensions) while preserving 95%+ similarity accuracy, improving K-NN query efficiency
- Integrated algorithm into HNSW indexing system and conducted performance benchmarking using OpenAI embedding datasets (100K+ vectors)
- Authored internal technical reports including statistical analysis and algorithmic benchmarking

Projects

TerraCast - Databricks x UW Hackathon - 1st Place | *University of Washington* | Devpost Link — Feb 2026

- Designed and led an end-to-end LightGBM ML pipeline on Databricks using Spark and Python for county-level corn and soybean yield prediction; applied yield detrending to isolate pure weather signal and drove EDA-based feature engineering on NOAA weather and regional datasets
- Identified and eliminated data leakage sources; implemented z-score-based anomaly detection to flag yield deviations relative to expected weather conditions with targeted visualizations
- Integrated Databricks Genie (LLM-powered AI assistant) for natural-language querying of model predictions via pandas DataFrame outputs, enabling non-technical stakeholder access

NFL Player Trajectory Prediction - Framework Comparison | *Kaggle* | Github Link — Oct 2025 - Dec 2025

- Implemented structurally equivalent GRU-based encoder-decoder models in PyTorch and TensorFlow for NFL player trajectory prediction, controlling for architecture, hyperparameters, and random seeds
- Conducted rigorous comparative analysis revealing significant autoregressive inference divergence despite structural equivalence (PyTorch RMSE: 1.55 vs TensorFlow: 19.47 on validation set)
- Authored technical report contextualizing empirical findings with established literature on exposure bias and error accumulation, documenting implications for deep learning framework reproducibility

WatchDawg - Seattle Crime Analytics Dashboard | *University of Washington* | Github Link — Sep 2025 - Dec 2025

- Developed interactive web dashboard analyzing 17+ years of Seattle Police Department crime data (500K+ incidents) with real-time filtering and geospatial visualization capabilities
- Engineered 11-stage data cleaning pipeline and memory-optimized architecture enabling stable deployment on resource-constrained environment (512MB RAM), reducing dataset noise by 35%
- Deployed production application on Render with 99%+ uptime; applied software engineering best practices including version control and modular pipeline design for maintainability